

ASSIGNMENT ON DATA CLEANING

Introduction

Data preprocessing cleans and prepares raw data, removing errors and inconsistencies. This step is important for improving accuracy and reliability in machine learning models.

Required Dataset

In this assignment we will use the “IBM HR Analytics Employee Attrition & Performance” dataset. You can download the dataset from the following link:

<https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>

Basic Description About the Dataset

The dataset is used to understand the factors that contribute to employee attrition (indicating whether an employee has left the company (Yes/No)) and to build models that can predict which employees are likely to leave.

The dataset consists of 35 columns (features) and 1470 rows (employee records).

Personal Attributes: Age, Gender, Marital Status, Education, etc.

Job-Related Attributes: Job Role, Department, Job Level, Job Satisfaction, etc.

Performance Metrics: Years at Company, Years in Current Role, Performance Rating, etc.

Compensation: Monthly Income, Stock Option Level, etc.

Attrition: The target variable, indicating whether an employee has left the company (Yes/No).

With the given dataset, you need to perform certain tasks which are given below. Remember that these steps are always crucial in dealing with any machine learning project. So we expect you to understand all these steps and implement them yourselves.

Required Tasks

- a. Understanding the dataset:
 1. Import the dataset in a notebook environment with python pandas.
 2. Show the number of attributes (columns) and number of records (rows).
 3. Show the statistics of the dataset (column wise mean, standard deviation, max, min etc.).
 4. Count the number of missing values in the dataset and show them.
 5. Count the number of duplicate values in the dataset and show them.

- b. Data Cleaning:
 1. If you find any missing values in the dataset (nan values) replace those with the column-wise median for numeric data, and with mode for categorical data.
 2. If you find any duplicates in the dataset, keep just one copy of the data.
 3. Remember, if any row in the target column (Attrition) is missing, you must drop that row.
 4. Remove any outliers present in the numeric columns.
 5. Standardize the string columns data to lower case.

- c. Creation of Input and Output Features:
 1. You need to split the data into two parts. The “Features” variable will consist of all the columns in the dataset except the target column. And the “Labels” variable will contain only the target column. This way, you can prepare the dataset for any machine learning model.
 2. The target column contains two classes (Yes/No). Convert them to numeric classes: 0 (No) and 1 (Yes) and replace them in the “Labels” column. This allows classification ML models to compute loss functions effectively.